



COMPUTER SECURITY

Lecture 6: Authentication Procedures and Database Security

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Authentication procedure

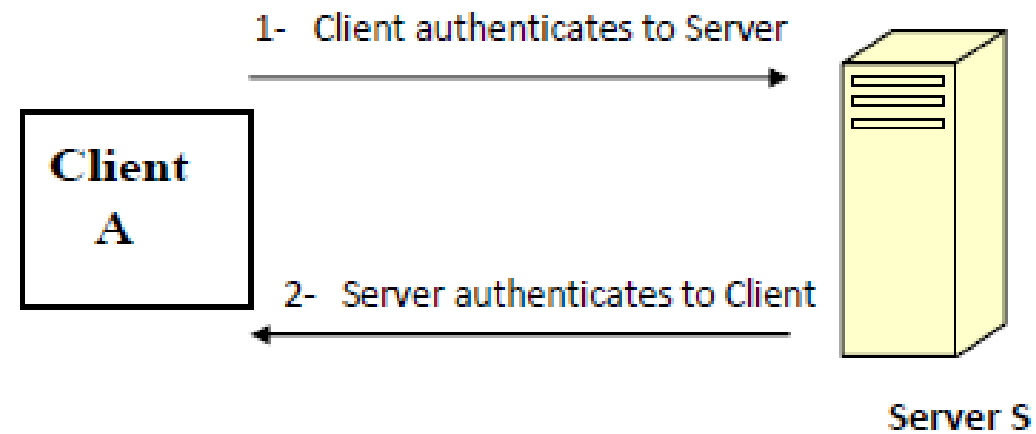


Figure 2: Two –Party Authentication

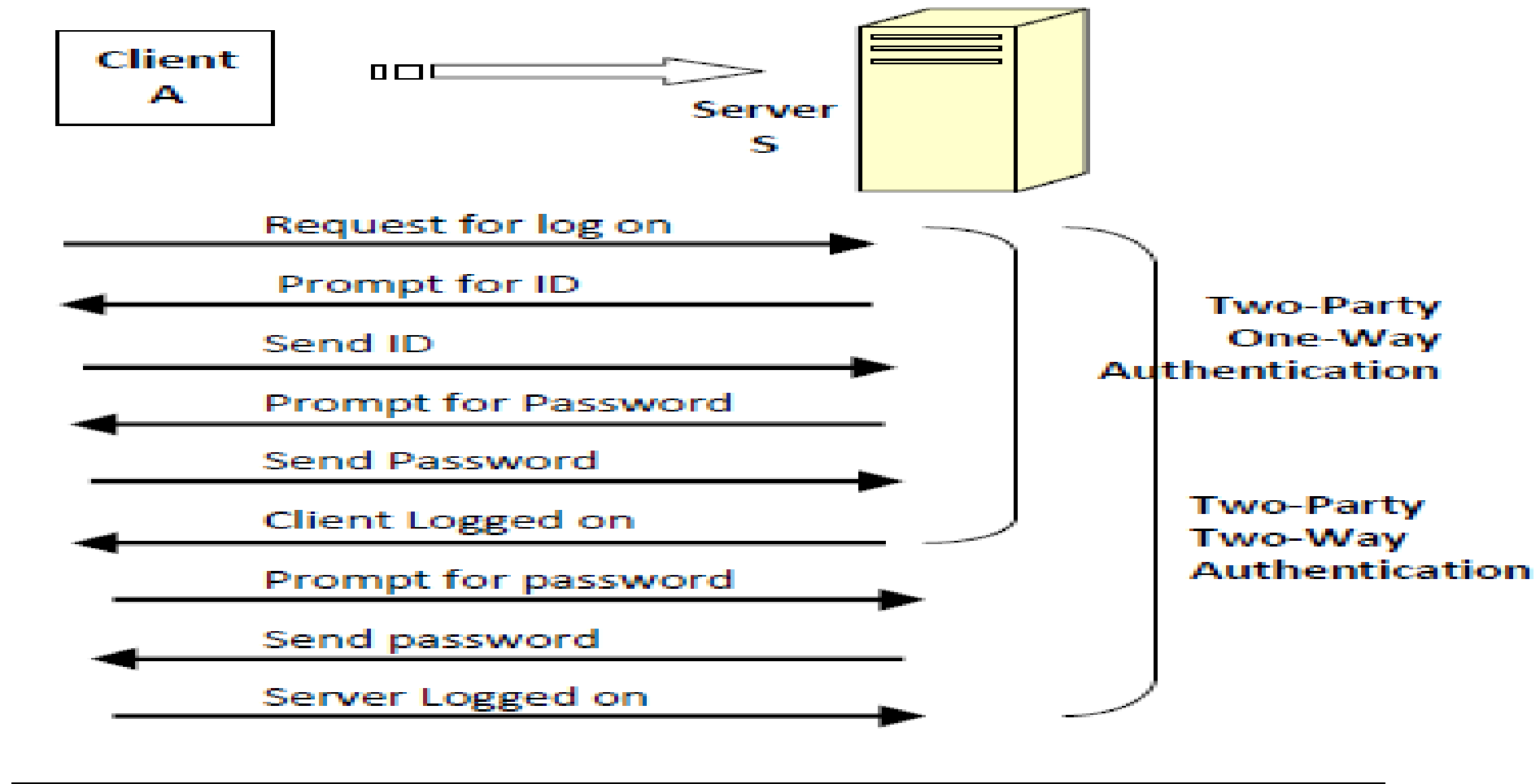


Figure 3: Two-Party One-Way Authentication and Two-Way Authentication.

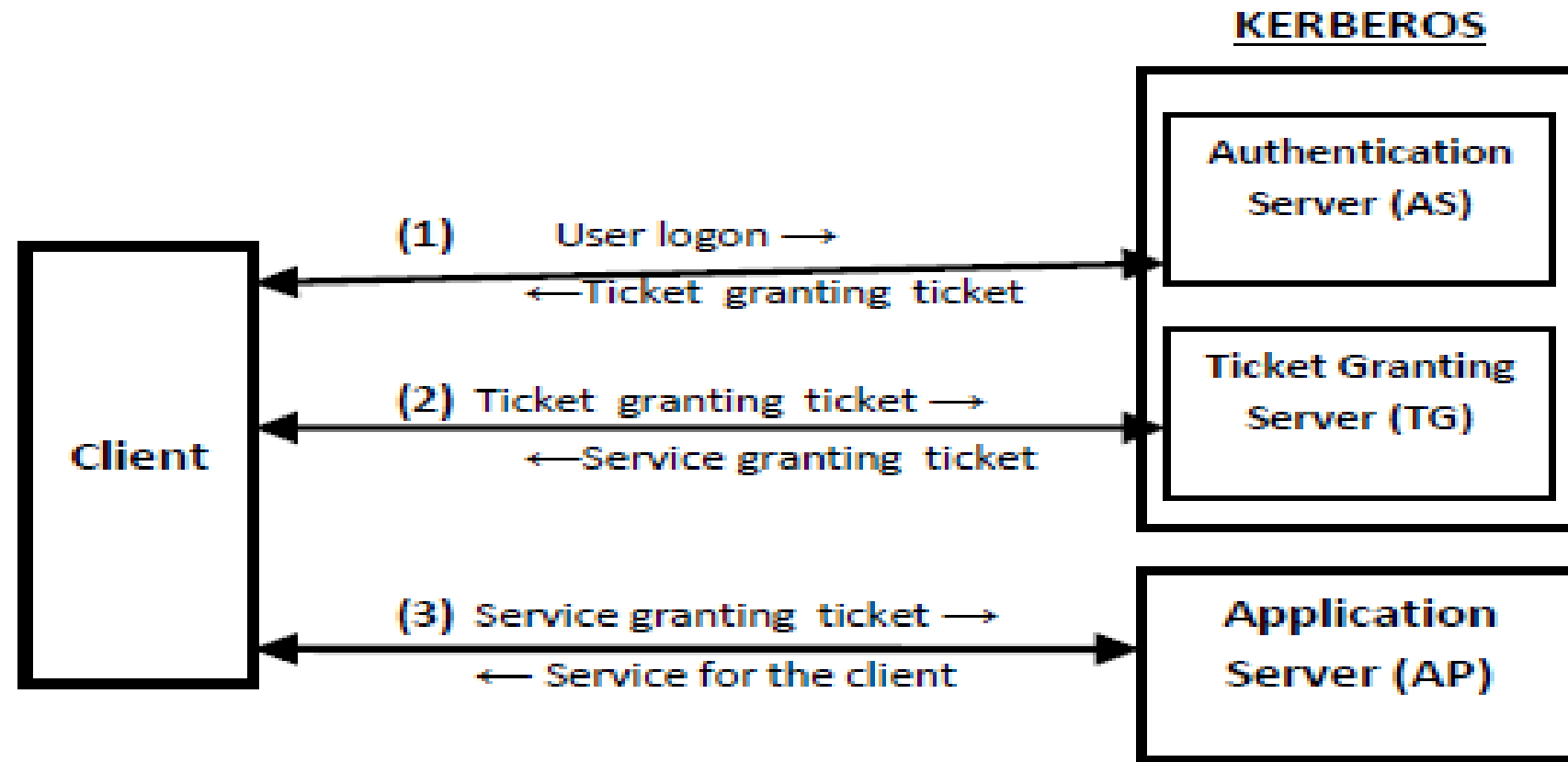


Figure 4: KERBEROS system

DB Security Requirements

- ***Physical DB integrity***. The data of a DB are immune to physical problems, s.a. power failures, and someone can reconstruct the database if it is destroyed through a catastrophe.
- ***Logical DB integrity***. The structure of the DB is preserved.
- ***Element integrity***. The data contained in each element are accurate.
- ***Audit ability***. It is possible to track who or what has accessed (or modified) the elements in the DB.

DB Security Requirements (cont.)

- **Access control.** A user is allowed to access only authorized data, and different users can be restricted to different modes of access (such as read or write).
- **User authentication.** Every user is positively identified, both for the audit trail and for permission to access certain data.
- **Availability.** Users can access the DB in general and all the data for which they are authorized.

Integrity of the DB

- Users must be able to trust the accuracy of the data values.
- DB admin. must be assured that updates of data are performed only by authorized individuals.
- Two situations can affect the integrity of a DB:
 - The whole DB is damaged.
 - Individual data items are unreadable.

Integrity of the DB (cont.)

- Integrity of the DB as a whole is the responsibility of the DBMS, OS, and the system manager.
- Periodic backups can be adequate controls against catastrophic failure.
- DBMS log files can be adequate controls to be able to restore the systems to a stable.

Element Integrity

- Elements integrity is their correctness or accuracy.
- Authorized users are responsible for entering correct data into DB.
- Programs must perform correct actions and produce correct results.
- Unfortunately, users and programs make mistakes.
 - DBMS should help to catch errors as they are made and to correct them after they are happened.

Element Integrity (cont.)

DBMS can apply corrective actions s.a.:

- Field checks, s.a. data type, range values, required property, ...etc
- Access control, who can access and modify the data elements without conflicting, duplication, ...etc.
- Maintaining a change log for the DB that contains both original and modified values.

These logs can be used to undo any change done by error.

Auditability

- For some applications it may be desirable to generate an audit record of all access (read or write) to a database.
- These records can help to maintain the database's integrity, or at least to discover after the fact who had affected which values and when.

Access Control

- The database administrator specifies who should be allowed access to which data, at the view, relation, field, record, or even element level. Also, specifies modes of access (read, write, delete, and append).
- Granularity size affects the efficiency of processing

Sensitive Data

DB can contain:

- 1- None of the data are sensitive s.a. a public library catalog database.
- 2- All of the data are sensitive s.a. defense-related databases.
- 3- Some but not all of the elements in the database are sensitive.

- The first two cases are easy to handle by access controls to the DB as a whole. Someone either is or is not an authorized user.
- The third one is the difficult case. DBMS must maintain access modes for each authorized user on different levels of the DB (tables, rows, columns, and data items). To simplify the job, privileges can be granted to the users as groups.

Sensitive Data (cont.)

Data can be sensitive, but so can their characteristics. Types of disclosure are:

- **Exact Data**
- **Bounds**
- **Negative Result**
- **Existence**
- **Probable Value**

Types of data disclosure

- Exact data
- Bounds
- Sensitive value is smaller than H , but bigger than L
- Might iteratively decrease range (binary search)
- Negative result
- Knowing that a person does not have zero felony convictions is sensitive, even if actual number is hidden
- Existence
- Knowing of existence of some data can be sensitive
- Probable value
- Sensitive data has value x with probability y

Inference

Inference is a way to infer or derive sensitive data from non-sensitive data. e.g. if the tables below summarizes the count of student and the pass students in such course, then if Lila is a CS student we can conclude that she fails; and other conclusions can be made.

Dept.	M	F
CS	3	1
CIS	4	3
MIS	2	6

Dept.	M	F
CS	2	0
CIS	2	3
MIS	0	3

THANK YOU

